

Vectors and Vector-Valued Functions



Vector operations in 3D

- 1 Let $\mathbf{a} = \langle 2, -1, 3 \rangle$ and $\mathbf{b} = \langle 1, 4, -2 \rangle$. Find:

a $\mathbf{a} + \mathbf{b}$

b $\mathbf{a} \cdot \mathbf{b}$

c $|\mathbf{a}|$

- 2 Find the cross product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 1, 0, 2 \rangle$ and $\mathbf{b} = \langle 3, 1, -1 \rangle$.

Vector-valued functions: velocity and acceleration

- 3 A particle's position is $\mathbf{r}(t) = \langle t^2, 3t, t^3 \rangle$.

a Find the velocity vector $\mathbf{v}(t)$.

b Find the acceleration vector $\mathbf{a}(t)$.

c Find the speed $|\mathbf{v}(1)|$.